

Bone tissue of ossification

1. Introduction

Bone tissue, like cartilage tissue, is a "skeletal tissue", a **specialized connective tissue**, characterized by the solid nature of the extracellular matrix (ECM).

The bone matrix has the particularity of **calcifying**, which makes it opaque to X-rays and allows the study of bones by radiography.

Our skeleton is essentially made up of pieces of bone tissue which, together with skeletal striated muscles, make up the locomotor system.

The skeleton has 3 functions:

- **Mechanical function:** bone tissue is one of the strongest tissues in the body, capable of withstanding mechanical stresses, giving bone its role in supporting the body and protecting organs.
- **Metabolic function:** bone tissue is a **dynamic tissue**, constantly remodeled under the effect of mechanical pressures, resulting in the release or storage of mineral salts, and thus ensuring to a large extent (in conjunction with the intestine and kidneys) the control of phosphocalcic metabolism.
- **Hematopoietic function:** bones contain in their medullary spaces the hematopoietic marrow, whose stem cells, which give rise to the 3 lineages of blood cells, are located in the vicinity of bone cells.

2. Embryological origin

Bone tissue, like connective tissue, derives from the **intra-embryonic mesoblast**.

The osteoformative cells that are responsible for the synthesis of the extracellular matrix of bone (bordering cells, osteoblasts and osteocytes) derive from **pluripotent mesenchymal stem cells**.

Osteoclasts are derived from the CFU-M **monocytic hematopoietic lineage** (monocyte blood lineage).

3. Histological structure

3.1 Bone cells (Fig. 01)

Bone tissue contains **4 types of cells:**

- Bordering cells, osteoblasts, and osteocytes are the osteoforming cells. Osteoclasts are osteoresorbing cells.
- Osteoblasts, osteoclasts and bordering cells are found on the surface of bone tissue, while osteocytes are located inside the bone matrix.

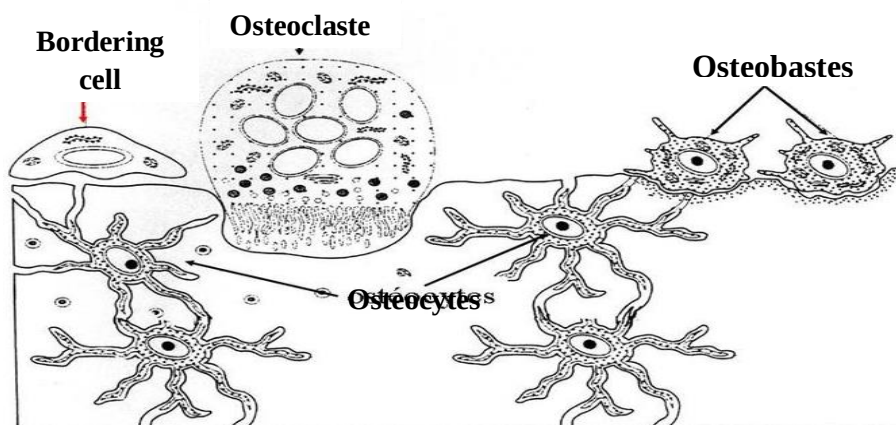


Figure. 01 The different types of bone tissue.

3.1.1 Osteoblasts (Fig.02)

- **cuboidal** osteoformative cells located on the external and internal **surface** of growing bone tissue. They are connected to each other and to osteocytes by gap junctions.
- Their plasma membrane contains abundant *alkaline phosphatase*. Osteoblasts produce the organic components of the ECM; therefore,
- Their cytoplasm is rich in organelles involved in protein synthesis (abundant granular endoplasmic reticulum, large Golgi apparatus).
- Osteoblasts can be degraded via **three pathways** :
 - transformation into osteocytes by becoming completely surrounded by ECM,
 - resting in the form of bordering cells lining the surfaces bony or
 - death by apoptosis.

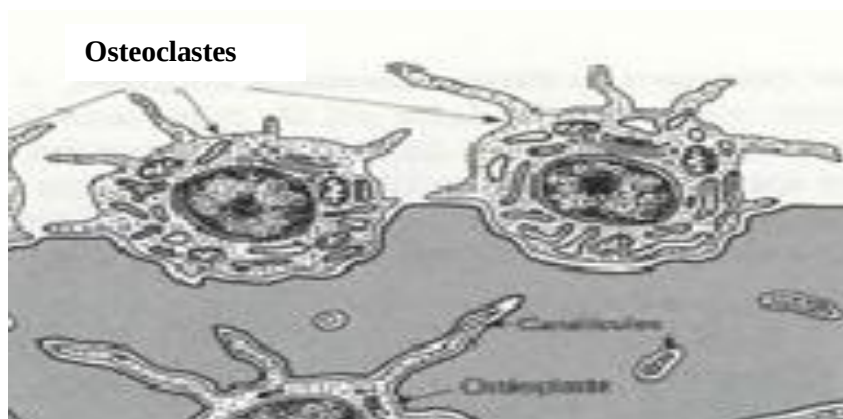


Figure.02 Osteoblast.

3.1.2 Osteocytes (Fig.03)

- These are differentiated osteoblasts, incapable of dividing, entirely **surrounded** by mineralized bone ECM.
- Osteocytes are located in locules (**osteoplasts**) from which anastomosing canaliculi extend, containing their thin, numerous, more or less long cytoplasmic extensions, connected to each other by gap junctions.
- Their cell body is smaller than that of osteoblasts, **fusiform**, and has fewer organelles than osteoblasts.
- Osteocytes, with limited synthetic capacities, participate in the maintenance of the bone matrix and contribute to the homeostasis of blood calcium.

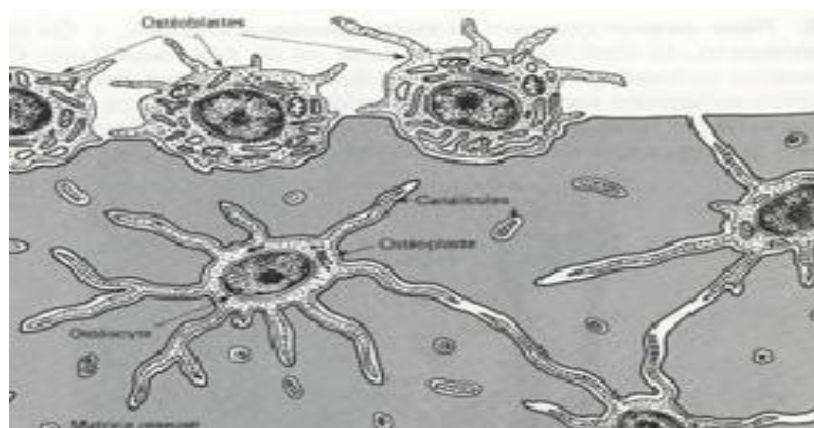


Figure.03 Osteocyte.

3.1.3 Bordering cells (Fig.04)

- ☛ The bordering cells are **resting osteoblasts**, which, if stimulated, can become active osteoblasts again.
- ☛ They cover bone surfaces which, at a given time, are not subject to **either** bone formation **or** resorption.
- ☛ **flattened and elongated** cells, possessing few organelles and connected to each other and to neighboring osteocytes by gap junctions.

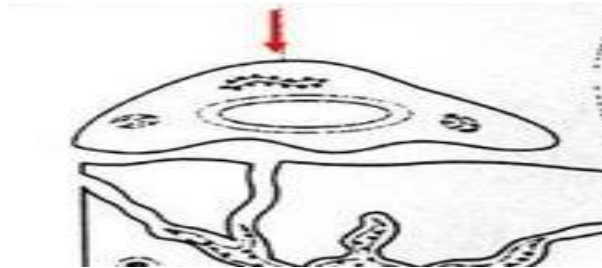


Figure.04 Bordering cell.

3.1.4 Osteoclasts (Fig.05)

- ☛ These are post-mitotic cells, very large, 20 to 100 μm in diameter, multinucleated, highly mobile, and capable of moving on the surface of bone trabeculae from one resorption site to another.
- ☛ When activated, the osteoclast, an osteoresorptive cell, develops its lysosomal apparatus and becomes strongly polarized; its plasma membrane differentiates into two domains separated by a tight ring of cell-ECM junctions:
 - **An apical domain** that develops a *brush border* in contact with the bone surface and
 - **A basolateral domain** located on the opposite side.

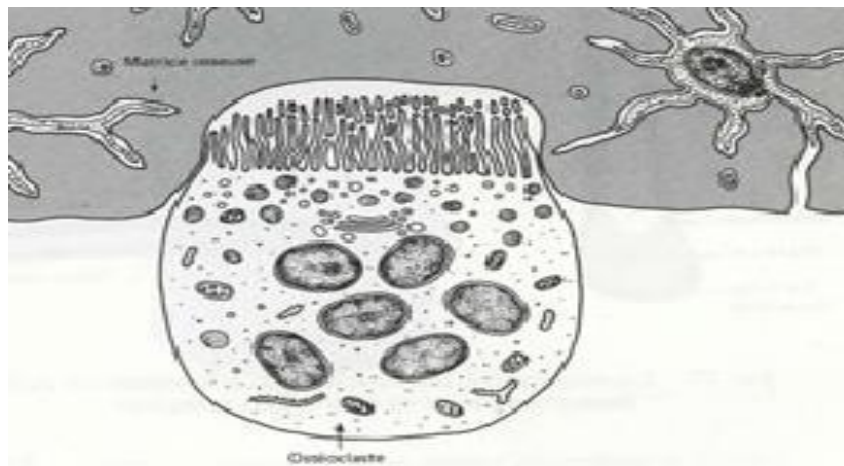


Figure.05 Osteoclast.

3.2 The extracellular matrix (ECM) of bone tissue is calcified

The ECM of bone comprises an organic part and a mineral phase.

3.2.1 The organic matrix: is composed

☺ From collagen I microfibrils, ☺ of proteoglycans, ☺ of osteopontin (linking hydroxyapatite to bone cells), ☺ of osteonectin (involved in mineralization through its affinity for collagen I and calcium), ☺ of osteocalcin (marker of mature osteoblasts,

involved in mineralization), ☺ of bone sialoprotein and ☺ of thrombospondin (allowing the attachment of bone cells to the ECM).

3.2.2 The mineral phase: is composed

☺ Of hydroxyapatite crystals (crystallized calcium phosphate) and ☺ of calcium carbonate.

☺ These crystals are visible under electron microscopy between and/or within collagen fibers, in the form of **small hexagonal needles**, electron dense.

☺ **The ions Ca^{++} and PO_4^{3-}** are located on the surface of the crystals participate in rapid exchanges with the interstitial fluid and therefore with the bloodstream.

☺ Bone, which contains 99% of the body's calcium, represents a calcium reservoir and plays a vital role in calcium and phosphorus metabolism.

4. Types of bone tissue (Fig. 06)

* **Lamellar bone tissue:** In adults, bone tissue is called **lamellar** because the bone matrix is arranged in superimposed **lamellae** where collagen microfibrils are arranged parallel to each other in a specific direction which changes in each successive layer.

* **Non-lamellar bone tissue:** In the fetus and young child, or in cases of **fracture**, or even during **certain diseases**, the collagen microfibril network produced by osteoblasts is irregular and the bone tissue is transiently **non-lamellar** ("tissue woven bone").

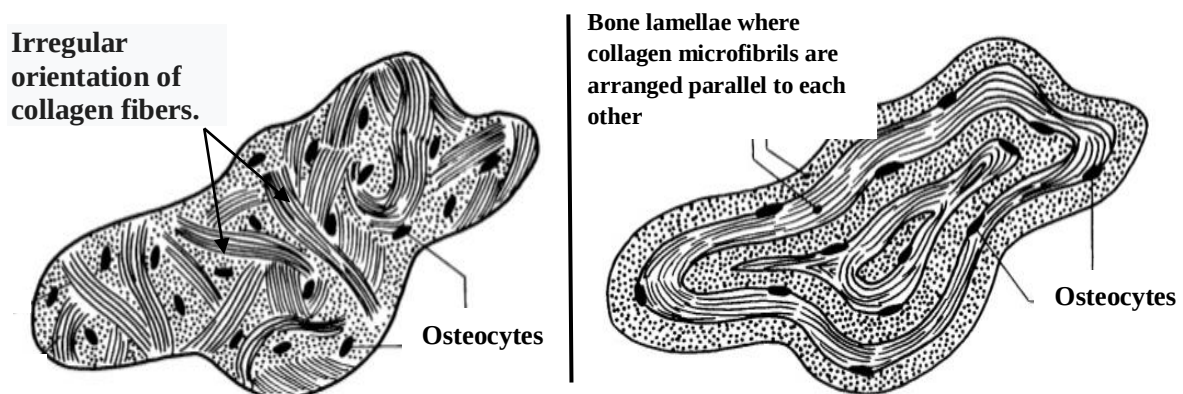


Figure.06 Types of bone tissue.

4.1 The three anatomical varieties of bone: long bones, short bones, flat bones.

Bones, primarily composed of bone tissue, also contain hematopoietic tissue, adipose tissue, vessels, nerves, cartilage tissue, and connective tissue.

♥ There are **3 anatomical varieties of bones**: long bones (like the tibia, the femur), short bones (like the carpal bones) and flat bones (like the sternum, the ribs).

♥ **The periosteum:** (Fig. 07)

The periosteum covers the external surface of all bones, except at the level of the articular cartilages, and is organized into two layers:

- An inner layer containing osteoprogenitor cells.
- An outer layer: rich in collagen fibers, some of which are grouped into bundles, penetrate the outer circumferential system and the interstitial system of the compact tissue and ensure close adhesion of the periosteum to the bone: **Sharpey's fibers**. This layer contains numerous blood vessels.

♥ **The endosteum:** it is a thin layer of connective tissue that lines the walls of all the vascular cavities of the bone tissue (Haversian canals, Volkmann's canals, medullary cavity of compact bone, cavities of spongy bone).

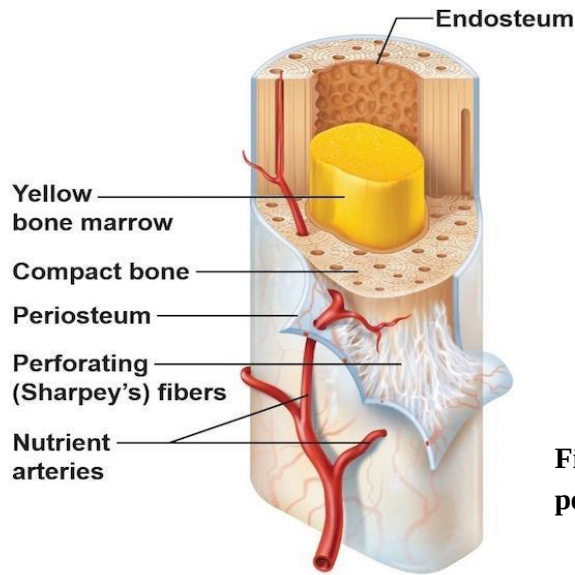


Figure.07 Endosteum and periosteum.

4.2 Lamellar bone tissue

4.2.1 Compact or dense Haversian bone tissue (Fig. 08)

- ☺ It consists of an assembly of cylindrical formations, called **Haversian systems** or **osteons**, arranged parallel to each other along the direction of the diaphysis. Each osteon travels a certain distance through the thickness of the diaphysis.
- ☺ In a cross-section, each osteon consists of 10 to 15 bone lamellae arranged concentrically around a canal approximately 50µm in diameter, which is the Haversian canal.
- ☺ Between the lamellae are the osteoplasts containing the cell bodies of the osteocytes.
- ☺ The **Haversian canal (H)** contains blood capillaries and unmyelinated nerve fibers embedded in a small amount of tissue loose connective tissue.
- ☺ The Haversian canals are connected to each other, to the medullary cavity and to the surface of the bone by transverse or oblique canals, **the Volkmann's canals (V)**.
- ☺ This arrangement gives the compact bone maximum strength.
- ☺ Between the osteons are bony lamellae, remnants of partially resorbed ancient osteons and constituting the **interstitial systems**.
- ☺ The diaphysis of long bones is bordered externally and internally by circumferential bony lamellae, creating **the external circumferential system** and **the internal circumferential system**.

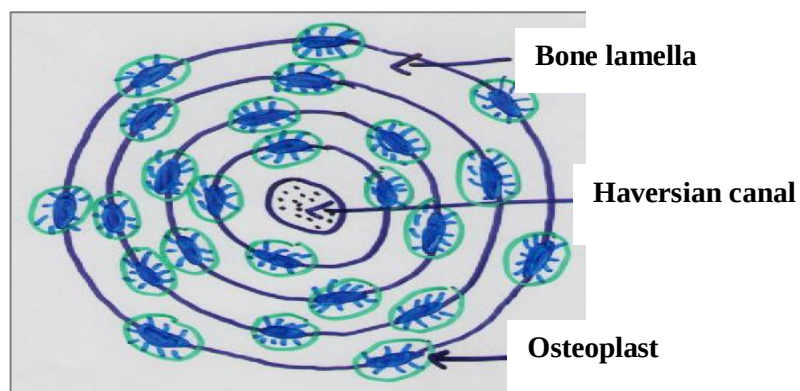


Figure.08 Osteon.

4.2.2 Spongy (or trabecular) bone tissue (Fig. 09)

♥ Spongy bone tissue is mainly located in short bones and flat bones (sternum, iliac wings) as well as in the epiphyses of long bones.

♥ It is formed by a three-dimensional network of spicules or trabeculae of bone tissue, branching and anastomosing, delimiting a labyrinth of intercommunicating spaces occupied by bone marrow and vessels.

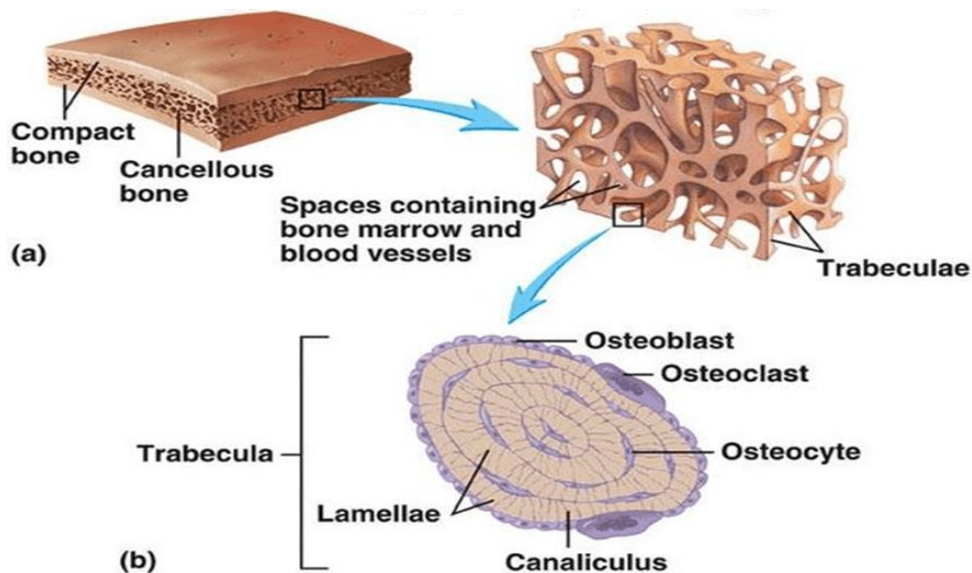


Figure.09 Trabecular bone tissue.

5. Vascularization of the bones

It is supplied by nutrient arteries and vascular branches coming from the periosteum: the periosteal branches.

6. Bone mass and bone loss

Until the age of 20, bone mass gradually increases. By this age, bone mass is established; it remains stable for a few years, then slowly declines with age. In women, bone loss accelerates significantly after menopause due to estrogen deficiency. This **osteoporosis** considerably increases the risk of fractures and most often justifies prolonged estrogen replacement therapy for women after menopause.

7. Pathologies

- ☛ **Osteopetrosis:** increased bone mass (absence of remodeling due to functional deficit of osteoclasts).
- ☛ **Osteoporosis:** due to hyperosteoclastosis, reduction of bone mass (imbalance of resorption/apposition; decrease in mineral density, fractures: femoral neck fracture is common in the elderly).
- ☛ **Osteomalacia :** (softening of bones and curvature, vitamin D deficiency), (**rickets** in children, **osteomalacia** in adults).
- ☛ **Paget's disease:** (excessive remodeling: bone deformities).
- ☛ **Osteoblast-derived tumors:**

- **Osteosarcoma:** is the most common **cancer** originating from bone cells; it derives from osteoblasts. It is more frequent in children and develops particularly in the bones around the knee.
- **Osteoid osteomas :** **benign** tumors, resulting in localized and painful swelling of a bone.

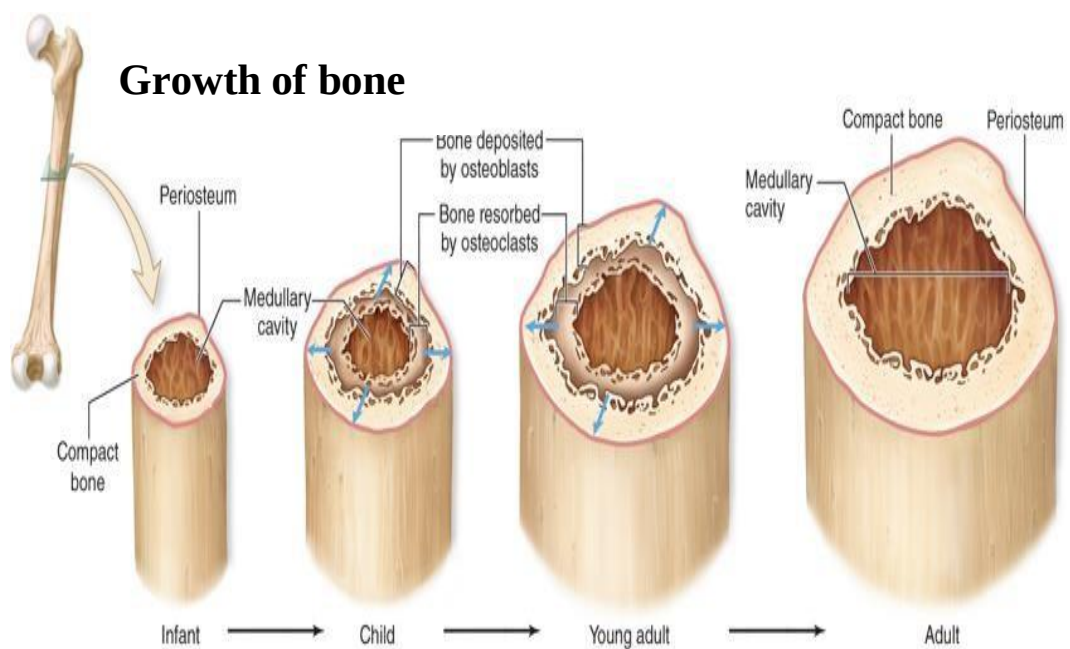
8. Histophysiology

☛ Calcium and phosphate reserve

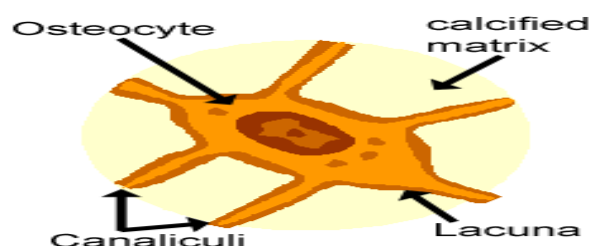
- * Bone tissue contains 99% of calcium and 85% of phosphorus.
- * Rapid and slow mobilization.
- * Parathyroid hormone (PTH) and calcitonin: PTH increases serum concentration in Ca^{2+} , calcitonin, has an antagonistic effect to that of PTH.

☛ Nutritional and hormonal effects on bone growth and remodeling

- * Growth hormone, **STH**, plays a role in the longitudinal growth of bone, through its action at the level of the growth plates.
- * **Sex hormones :** Estrogens and androgens stimulate production of the bone matrix.
- * Vitamin D: Vitamin D3 plays an important role in promoting absorption intestinal calcium and its fixation on bone.
- * Vitamin C.



- Bone formation and bone resorption go hand in hand during the growth of bone.



Ossification

1. Introduction

- It is a process of building bone tissue.
- Regardless of the type of bone tissue obtained (lamellar or non-lamellar), regardless of where the bone tissue forms (within connective tissue or cartilage), the formation of bone tissue, or **histogenesis**, always follows the same basic process:

We can distinguish four stages:

- The differentiation of osteoblasts from stem cells;
- The secretion of organic matrix by osteoblasts;
- The mineralization of the organic matrix, and finally;
- The arrival of osteoclasts.

2. The stages of ossification

Ossification occurs in several stages:

- **Primary ossification:**
 - from connective tissue (membrane ossification or endoconnective ossification).
 - from a cartilaginous rudiment (hyaline cartilage) (endochondral ossification).
- **Secondary ossification:** from already formed bone tissue (primary bone).
- **Tertiary ossification:** in adults. Corresponds to permanent bone remodeling.

2.1 Primary ossification

2.1.1 Endoconjunctival ossification (Fig.10)

This is the most basic and easiest mode to understand.

From connective tissue, fibroblasts (or mesenchymal cells) differentiate into osteoblasts, which produce a pre-osseous substance (osteoid), which is rapidly mineralized.

As soon as a small area of bone tissue is formed, other mesenchymal cells differentiate into osteoblasts, which arrange themselves all around the bone area, forming an "osteoid rim," and continue the same process, resulting in an appositional increase in bone tissue. And so, gradually, the bone tissue replaces the connective tissue.

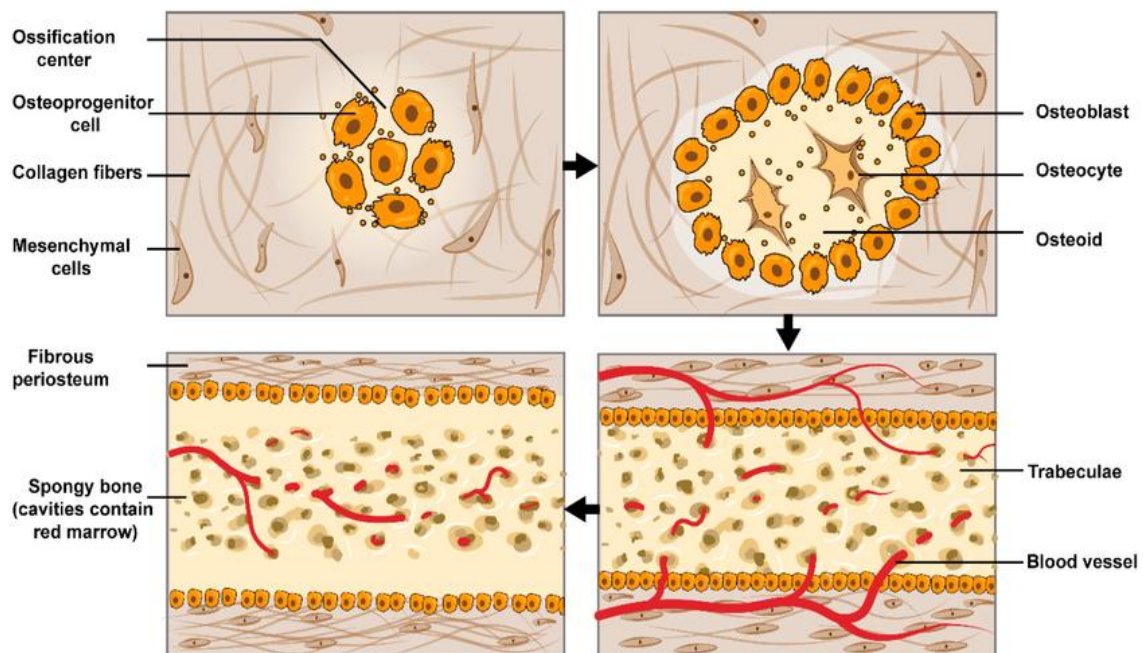


Figure.10 Endoconjunctival ossification.

2.1.2 Enchondral ossification (Fig.11)

Endochondral ossification concerns the bones located below the base of the skull, except for the clavicles.

It begins towards the end of the 2nd month of development from hyaline cartilage.

The phenomenon itself is more complicated because the formation of bone tissue is intertwined with the destruction of the cartilage tissue at which it takes place.

Enchondral ossification develops from a diaphyseal ossification center and then continues at the level of the two epiphyseal ossification centers.

This " **ossification center** " or " **ossification point** " is the site of events:

1- **The formation of a bony sheath or periosteal ring** around the hyaline cartilage model.

2- **Calcification of the center of the cartilage**

- The chondrocytes hypertrophy; their cytoplasm accumulates glycogen and becomes vacuolar; their chondroplast enlarges proportionally at the expense of the cartilaginous organic matrix, which is reduced to thin trabeculae.
- These thin trabeculae calcify by the deposition of hydroxyapatite crystals in the ground substance progressively, the chondrocytes degenerate and die, while the enlarged **chondroplasts** become confluent.

3- **Conjunctival-vascular bud**

- **Blood capillaries** proliferate and penetrate the open chondroplasts, which have become vacant due to the death of the chondrocytes.
- These capillaries carry with them undifferentiated mesenchymal cells.
- Some differentiate into **hematopoietic cells** and others, upon reaching contact with the cartilaginous trabeculae, differentiate into **osteoblasts**, arrange themselves on the surface of the residual trabeculae of calcified cartilaginous matrix and develop a layer of bone tissue against them.
- Formation of spongy bone.

4-**Lengthening of the diaphysis and formation of the medullary cavity:** during ossification, secondary ossification points appear in the epiphyses.

5- **The ossification of the epiphyses**

At the end of this stage, cartilage remains only in the articular cartilage and the growth plate cartilage.

- ☛ **At the level of the two epiphyses**, vascular penetration induces the formation of two epiphyseal ossification centers following the same steps described previously, resulting in the formation of a primary bone at the level of the two epiphyses.

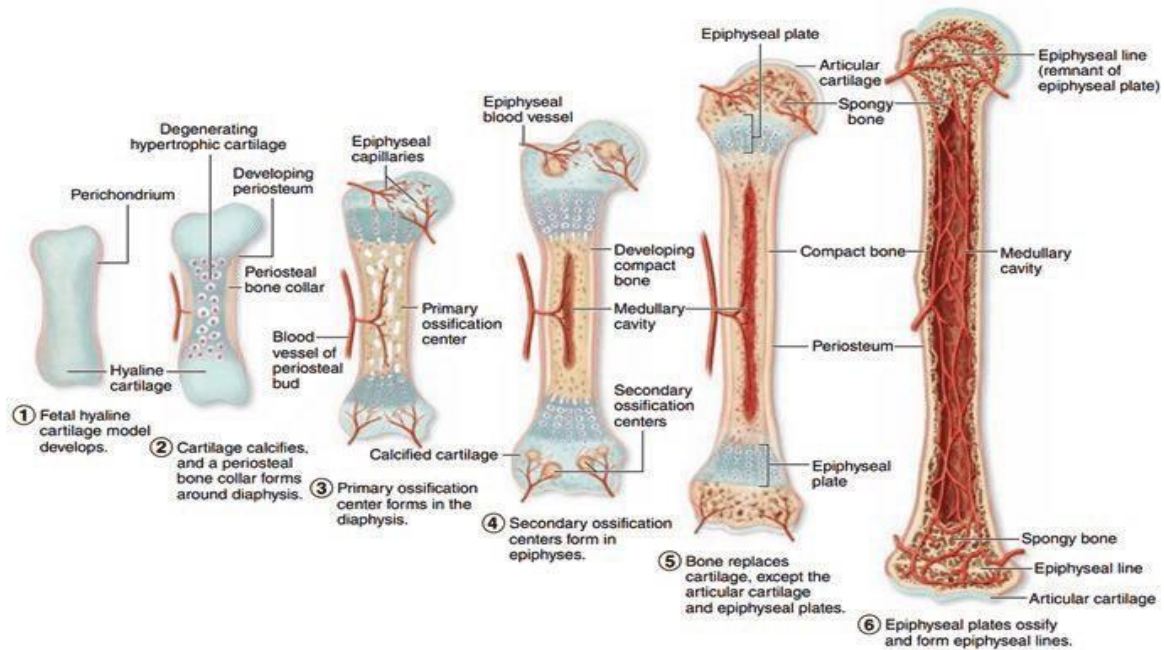


Figure.11 Endochondral ossification.

2.2 Growth of long bones

* The growth plate: (Fig. 12)

The growth of long bones Continuing throughout childhood and adolescence , bone lengthening is ensured by endochondral **ossification which** occurs at each end of the long bones, through the proliferation of **growth plates** .

Note: after puberty, chondrocytes stop multiplying, growth plates disappear: growth is complete.

* **Thickness growth: endoconnective** type **periosteal** ossification occurs through the successive peripheral apposition of new layers of bone tissue originating from the activity of the periosteum.

Organization of Cartilage within Epiphyseal Plate of Growing Long Bone

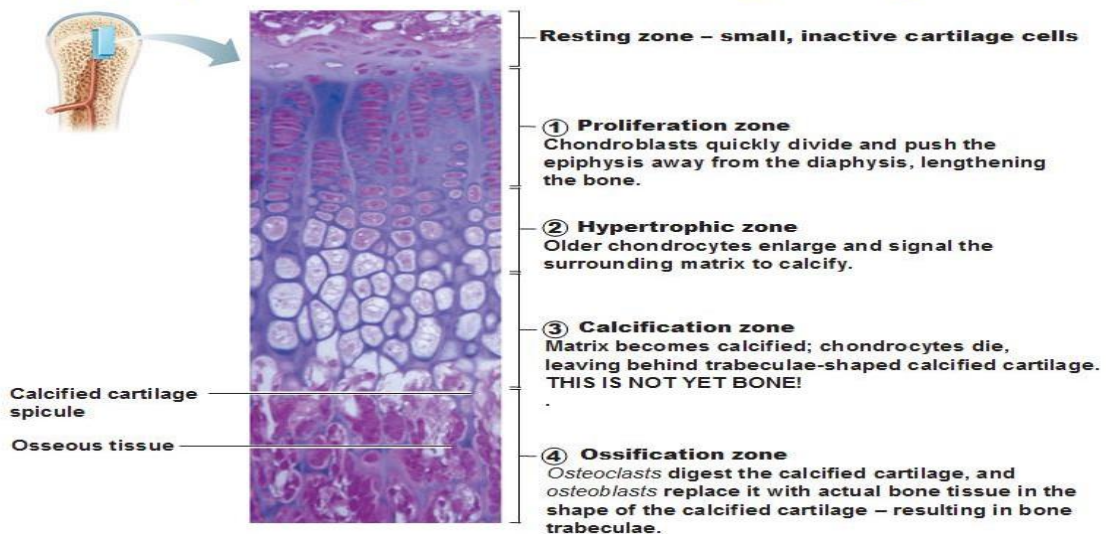


Figure.12 Growth plate.

2.3 Secondary ossification:

Secondary ossification mechanisms correspond to the phenomena leading to bone lamellation through successive osteoclastic resorption and osteoblastic reconstruction. They allow the replacement of pre-existing bone tissue (from primary ossification: non-lamellar, primitive, primary, immature bone) by new lamellar type bone tissue (secondary, mature): (compact bone tissue and spongy bone tissue).

2.4 Tertiary ossification or bone remodeling:

Includes all the mechanisms that modify the structure of lamellar bones through destruction and subsequent bone resynthesis, allowing for skeletal adaptation:

- To growth
- To maintain calcium and phosphate balance
- To new mechanical constraints that may appear during the life (increased weight, particular working conditions).
 - Bone is constantly renewing itself.
 - It is a continuous cycle of bone formation and breakdown, which is a living tissue.
 - Osteoclasts will first destroy the previously formed bone, and **osteoblasts will** then intervene to rebuild a new bone.
 - Bone remodeling, by constantly renewing the bone, allows it to be strong and resilient.

The remodeling process involves **4 phases**

- **Activation:** differentiation of pre-osteoclasts into osteoclasts.
- **Osteoresorption:** the degradation of bone ECM by osteoclasts.
- **Inversion:**
 - * Death of osteoclasts by apoptosis.
 - * Replacement by macrophages that clean up the debris.
 - **Bone formation of bone tissue** : synthesis of a non-mineralized ECM (osteoid) by osteoblasts followed by its mineralization.
 - **Rest or quietude.**

3. Repair after a fracture

At the fracture site, tissue destruction and hemorrhage (hematoma) are observed, which attract neutrophils and macrophages to the site to clean up tissue debris.

The damaged area will be filled successively by:

- Connective tissue;
- Cartilage tissue;
- From immature bone (primary ossification);
- From mature lamellar bone (bone callus: secondary ossification).